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GIS 4850

Mini-Project 1

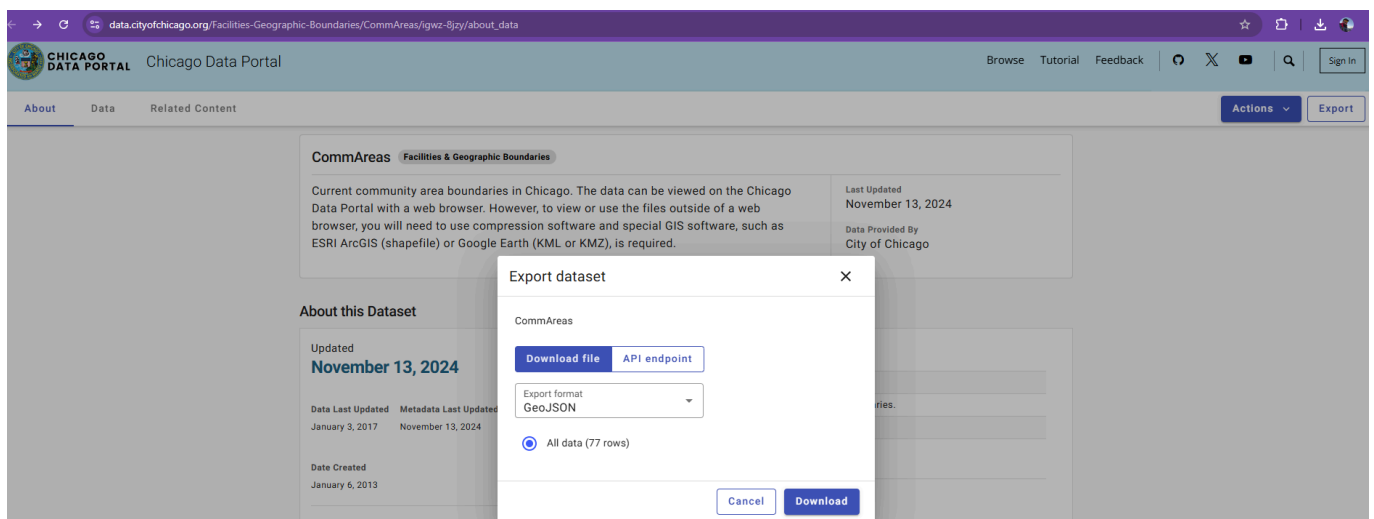
February 14, 2025

For In this tutorial, I will explain how to create a map that displays the average elevation for communities located in the Chicago, IL area. To do so requires acquiring raster elevation data of the area, vector data of the communities, and merging the two together to produce an understandable and intuitive map.

Download Shapefiles

It is essential that the data is gathered in the correct format. Incorrect formats will not load properly into ArcGIS Pro and won't be usable for this project.

1. Download shapefiles for Chicago community areas (**CommAreas**) from the Chicago Data Portal at <https://data.cityofchicago.org/> by clicking **Export** in the top right and then downloading the **GeoJSON** file format.

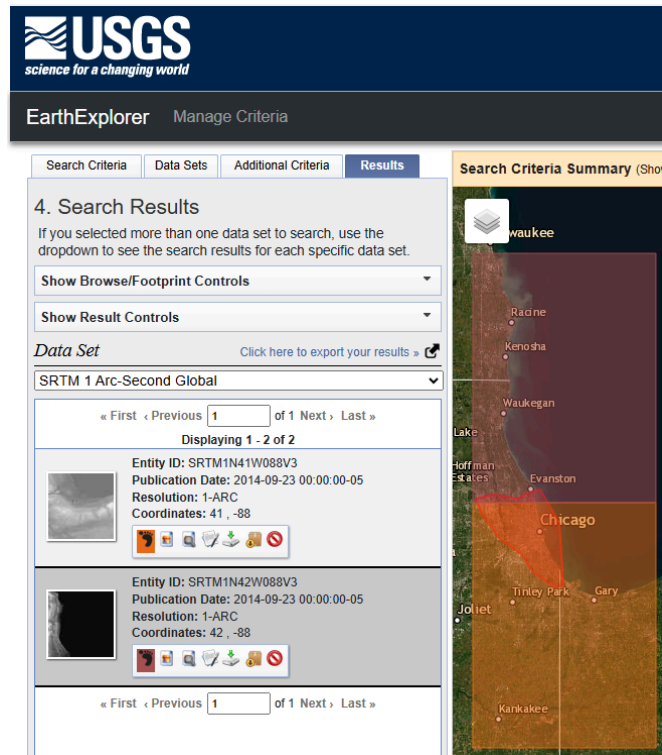


2. Extract data and unzip files with 7-Zip.
3. Create a new map named **Chicago Communities Average Elevation** and add a folder connection to the Chicago Community data.
4. Turn on labels for communities by selecting **CommAreas** in the Contents pane
5. Click the **Labeling** tab on the top ribbon and turn on labels using the button on the top right.
6. To display the names of communities, change the field to **COMMUNITY** in the drop down menu.
7. To make it easier to determine which label goes with which community, adjust the color of the font as well as its size to better fit within each polygon and make it easy to read.
8. If labels are placed in a manner that is not legible, convert labels to annotations to edit and move them.

Download Digital Elevation Models

In this case, the two DEMs were collected on the same day and line up perfectly. Later on, the two will be merged together as a mosaic. If they did not match so well, the Raster Calculator tool would be utilized to merge them together.

1. Download **SRTM** DEM raster data from the USGS EarthExplorer website, Entity ID **SRTM1N41W088V3** and **SRTM1N42W088V3**.



2. Extract data and unzip files with 7-Zip.
3. Make a folder connection to files.
3. Calculate statistics when adding to ArcGIS Pro Map.

Join Rasters - Create a Mosaic

In order to accurately calculate the average elevation for each community, the two DEM layers must be stitched together. This can be done using the Mosaic to New Raster tool. If not done properly, inaccurate data could result.

1. In the Contents Pane, change the layer name of **n41_w088_1arc_v3.tif** to **DEM_1_South** and **n42_w088_1arc_v3.tif** to **DEM_2_North**.
2. Join rasters using the **Mosaic To New Raster** geoprocessing tool.
3. The input rasters are **DEM_1_South** and **DEM_2_North**.
4. Ensure the output location is in the project database.
5. Set the “Raster Dataset Name with Extension” to **ChicagoDEM_Combined**.
6. The Spatial Reference for the raster should be **32 bit float**.
7. Enter **1** for the number of bands.
8. Leave other fields as they are.
9. Run the tool.

Geoprocessing

Mosaic To New Raster

Parameters Environments

Input Rasters

- DEM_1_South
- DEM_2_North

Output Location

Mini Project 1.gdb

Raster Dataset Name with Extension

ChicagoDEM_Combined

Spatial Reference for Raster

GCS_WGS_1984

Pixel Type

32 bit float

Cellsize

Number of Bands

1

Mosaic Operator

Last

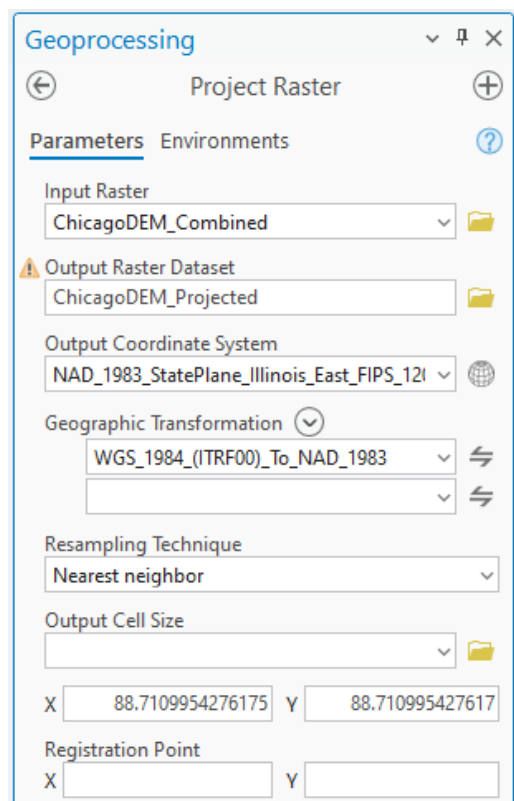
Mosaic Colormap Mode

First

Reproject Raster

In order to ensure all data lines up and produces accurate results, all layers must be in the same map projection. In this case, only the ChicagoDEM_Combined needs to be reprojected to match that of the CommAreas layer, which is the Illinois State Plane East projection.

1. Reproject the new raster created in the last step, **ChicagoDEM_Combined**, to the Illinois State Plane using the **Project Raster** geoprocessing tool.
2. The input raster is **ChicagoDEM_Combined**
3. Name the Output Raster Dataset **ChicagoDEM_Projected**
4. Set the output coordinate system to **NAD_1983_StatePlane_Illinois_East_FIPS_1201_Feet**, which can be done more easily by selecting the **CommAreas** vector layer in the drop down menu that is already in the Illinois State Plane projection.
5. Leave the resampling technique to nearest neighbor.
6. Run the tool.



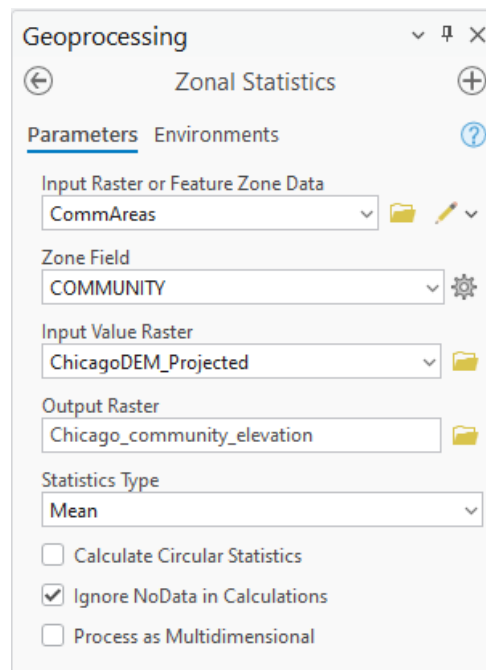
The screenshot shows the 'Project Raster' tool window in a GIS application. The 'Parameters' tab is active, displaying the following settings:

- Input Raster:** ChicagoDEM_Combined
- Output Raster Dataset:** ChicagoDEM_Projected
- Output Coordinate System:** NAD_1983_StatePlane_Illinois_East_FIPS_1201_Feet
- Geographic Transformation:** WGS_1984_(ITRF00)_To_NAD_1983
- Resampling Technique:** Nearest neighbor
- Output Cell Size:** (empty)
- Registration Point:** X: 88.7109954276175, Y: 88.710995427617

Compute Average Elevation for Each Community

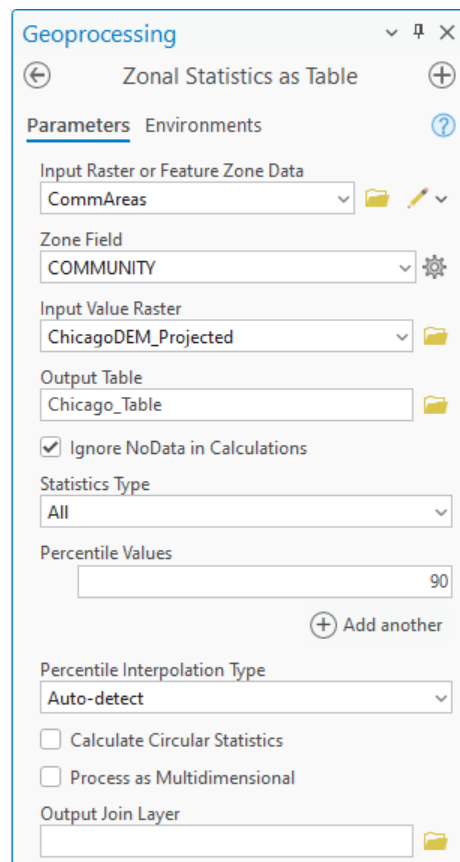
In this step the DEM layers will be merged with the vector layers to produce a result calculating the average elevation for each of the communities.

1. Open the **Zonal Statistics** geoprocessing tool.
2. Use **CommAreas** as the Input Raster or Feature Zone Data.
3. Set **ChicagoDEM_Projected** as the Input Value Raster.
4. Choose **Community** as the Zone Field.
5. Select **Mean** as the Statistics Type.
6. Name the Output Raster **Chicago_community_elevation**.



Using Zonal Statistics as a Table

1. Open the **Zonal Statistics as Table** geoprocessing tool
2. Select **CommAreas** as the Input Raster or Feature Zone Data.
3. Choose **ChicagoDEM_Projected** as the Input Value Raster.
4. Use **COMMUNITY** as the Zone Field.
5. Select “All” for the Statistics Type.
6. Name the Output Table **Chicago_Table**
7. Run the Tool



The screenshot shows the 'Zonal Statistics as Table' tool interface. The 'Parameters' tab is active. The 'Input Raster or Feature Zone Data' is set to 'CommAreas'. The 'Zone Field' is set to 'COMMUNITY'. The 'Input Value Raster' is set to 'ChicagoDEM_Projected'. The 'Output Table' is set to 'Chicago_Table'. The 'Ignore NoData in Calculations' checkbox is checked. The 'Statistics Type' is set to 'All'. The 'Percentile Values' list contains the value '90'. The 'Percentile Interpolation Type' is set to 'Auto-detect'. The 'Calculate Circular Statistics' and 'Process as Multidimensional' checkboxes are unchecked. The 'Output Join Layer' field is empty.

Geoprocessing

Zonal Statistics as Table

Parameters Environments

Input Raster or Feature Zone Data
CommAreas

Zone Field
COMMUNITY

Input Value Raster
ChicagoDEM_Projected

Output Table
Chicago_Table

☒ Ignore NoData in Calculations

Statistics Type
All

Percentile Values
90
+ Add another

Percentile Interpolation Type
Auto-detect

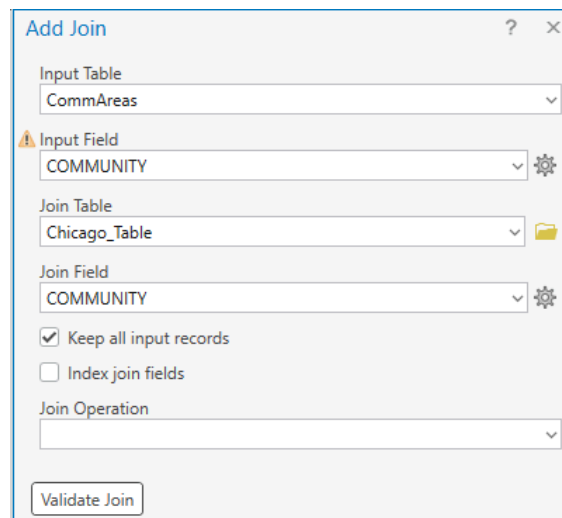
☐ Calculate Circular Statistics
☐ Process as Multidimensional

Output Join Layer

Join the Table to the Communities Layer

If the table is not correctly joined to the CommAreas layer, the data gathered from the DEM layer will not be properly represented in the vector layers of communities.

1. Right click the **CommAreas** layer in the Contents window
2. Highlight **Joins and Relates** -> **Add Join**
3. The Input Table is **CommAreas**, Input Field is **COMMUNITY**, Join Table is **Chicago_Table**, and the Join Field is **COMMUNITY**. Click **OK**.

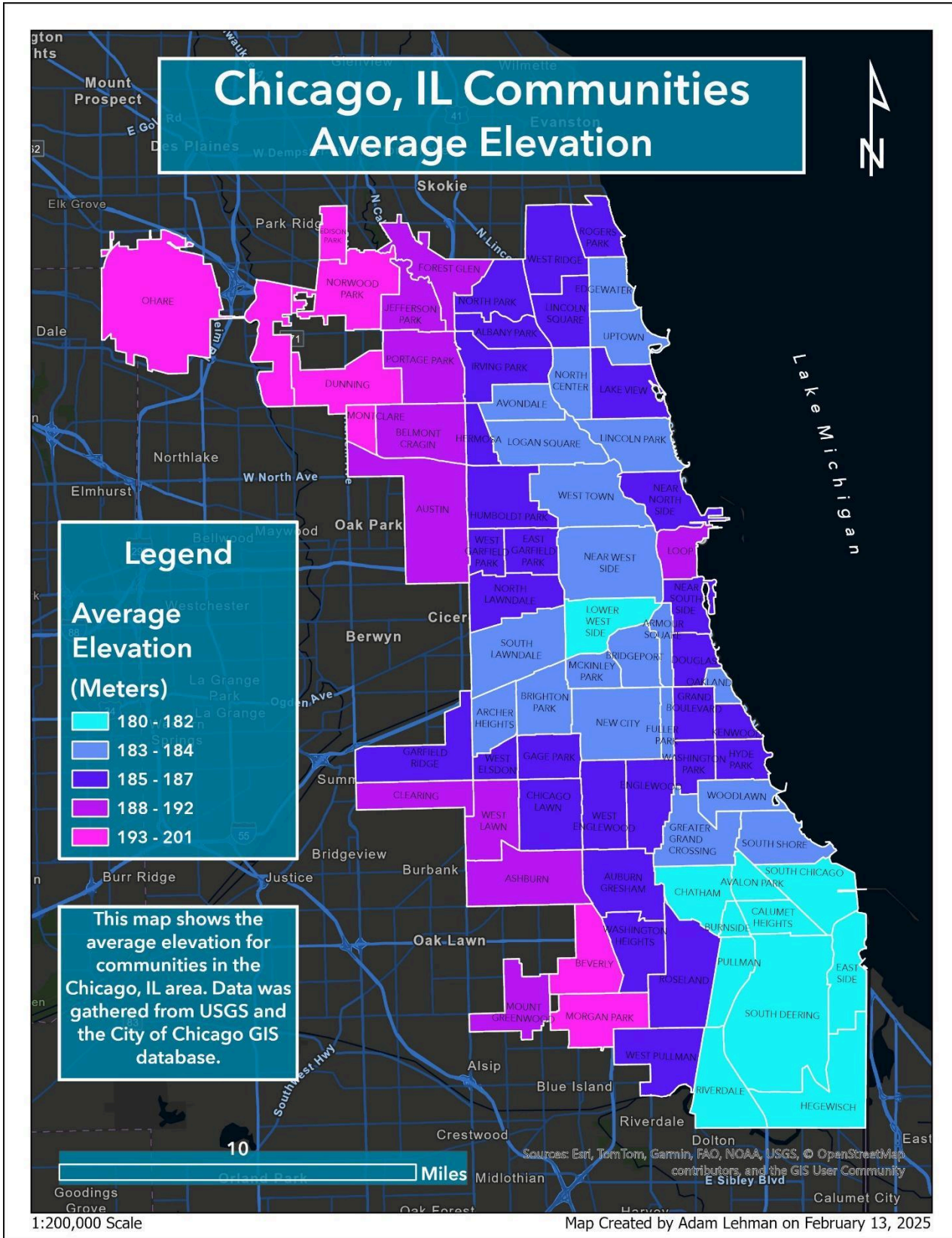


5. Change the name **CommAreas** in the Contents Pane to **Average Elevation**.
6. Open the symbology pane for the **Average Elevation** layer and change the field to **MEAN**, the method to **Natural Breaks (Jenks)**, and select a color scheme that clearly communicates the data.

Create Map Layout

1. Select the **ANSI Portrait - Letter - 8.5" x 11"** (or whichever size is preferred) map layout option from the drop-down menu under the **Insert** ribbon at the top of the screen.
2. Create a layout that is visually appealing, communicates the information well, and includes a title, legend, scale bar, north arrow, and attributes/credits.
3. Export map and save.

Final Product



Sources

Chicago, City of. "CommAreas: City of Chicago: Data Portal." *Chicago Data Portal*, 3 Jan. 2017, data.cityofchicago.org/Facilities-Geographic-Boundaries/CommAreas/igwz-8jzy/about_data.

Survey, USGS - U.S. Geological. "Earthexplorer." *EarthExplorer*, earthexplorer.usgs.gov/. Accessed 12 Feb. 2025.